MIL-F-12716A(Ord)
7 March 1960
SUPERSEDING
MIL-F-12716 (Ord)
13 May 1953

MILITARY SPECIFICATION

FILTERS, LIGHT, TELESCOPIC INSTRUMENT: LAMINATED

1. SCOPE

1.1 Scope.- This specification covers laminated filters used with optical fire control instruments. Filters shall consist of a sheet of colored plastic cemented between two colorless glass plates.

1.2 Classification.- Filters shall be of the following types, classes, and grades:

Type I - Amber (See Figure 1)
Type II - Red (See Figure 2)
Type III - Neutral, in 3 classes as follows:

Class 1 - (See Figure 3)
Class 2 - (See Figure 4)
Class 3 - (See Figure 5)

Grade A - For use near telescope objective lenses.
Grade B - For use near the focal plane of an eyepiece.
Grade C - For use with the unaided eye.

1.2.1 Cross reference.- For cross references to types, classes, and grades of Specification MIL-F-12716, see 6.5.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools: Packaging of

FSC 1240

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking for U. S. Military Property

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one filter representative of production, shall be made available to the Government to determine if the filter meets all the requirements of this specification.

3.2 Materials.- Materials shall be in accordance with applicable drawings and specifications forming a part of this specification.

3.2.1 Glass.- Glass used for the filter plates shall be as specified on applicable drawing.

3.2.2 Plastic material.- Plastic material shall be free from tears, interruptions, wrinkles, undulations, or opaque inclusions which would impair the optical properties of the filter.

3.3 Construction.- The filter shall be manufactured in accordance with the applicable drawings pertaining thereto. Where no drawing exists, the size shall be specified in the contract (see 6.2). The requirements of this specification are detailed only to the extent considered necessary to insure the inherent quality and the performance capabilities of the product prior to final acceptance by the Government.

3.4 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-O-13830:

(a) Cleanliness
(b) Surface qualities

3.5 Environmental.

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3.5.1 Storage temperature.- The filter shall show no evidence of physical failure and shall meet the requirements of this specification at standard ambient temperature (60 to 90 degrees Fahrenheit) after being exposed to ambient temperatures of plus 160 and minus 65 degrees Fahrenheit.

3.6 Cemented surfaces.- The filter shall show no evidence of discolorations, bubbles, or separations more than 2 millimeters from the edge.

3.7 Optical characteristics.

3.7.1 Uniform light transmission.- The filter shall show uniform light transmission over its entire area when examined in natural light.

3.7.2 Percent light transmission.- The optical characteristics of all grades of filters shall conform to the requirements of the appropriate type and class as shown on Figures 1 to 5 inclusive. The graph on each figure indicates the special transmission curves corresponding to the maximum and minimum limits for each type and class of filter.

3.7.3 Angular deviation.- The assembled filters shall not produce an angular deviation greater than:

Grade A filters: 1 minute of arc
Grade B filters: 5 minutes of arc
Grade C filters: 5 minutes of arc

The above requirements shall apply to all types and classes in accordance to grade indicated.

3.7.4 Image quality.

3.7.4.1 Grade A.- The resolution of the telescopic system shall not be impaired and there shall be no evidence of parallax between target image and telescope crosslines when an assembled grade A filter is placed in front of an objective system of 6 power and adjusted for infinity with an objective aperture at least as large as the usable portion of the filter.

3.7.4.2 Grade B.- The filter shall not impair the image quality when a grade B filter is placed in front of a 10 power eyepiece having a free aperture of not less than 15 millimeters and tested in the geometrical center and in four regions at the edge.

3.7.4.3 Grade C.- The positive or negative refractive power of a grade C filter shall not exceed 1/8 diopter in any meridian, and the difference in astigmatism of the filter in any two meridians shall not exceed 1/8 diopter.

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3.8 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

(e) Measures for preventing any significant departure of product quality from standards.

4.1.1.1 The contractor will not be restricted to the inspection station or to the method of inspection listed, provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and

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(b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered, shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Production model.

4.2.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

4.2.1.1 These tests shall be conducted at the contractor's facility or Government approved laboratory using test equipment designed or procured for use on the remaining items.

4.2.1.2 The model shall meet all the requirements of this specification using the inspection method prescribed herein.

4.2.2 Disposition of rejected production model.- Should the production model fail to pass the test for Government acceptance, the contractor shall cease those operations which are causing rejectable material and make the necessary corrections. When corrections have been made, a new production model shall be tested by the contractor under the surveillance of the Government. Operations causing rejections shall not be resumed until a production model has successfully passed all inspections specified.

4.3 Inspection provisions.

4.3.1 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.3.2 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

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4.3.3 Acceptance tests.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements of telescopic instrument light filters to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.3.4 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL value is expressed in percent defective

MAJOR: AQL 4.0%	Req't Par.	QA Par.	Test Method
Storage temperature	3.5.1	4.5.1	Standard Measuring Equipment (SME)

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL value is expressed in percent defective

MAJOR: AQL 1.0%	Req't Par.	QA Par.	Test Method
1. Cleanliness	3.4	4.5.2	MIL-O-13830 Visual
2. Surface quality	3.4	4.5.3	MIL-O-13830 Surface Quality Standard 7641866
3. Cemented surfaces	3.6	4.5.4	Visual SME
4. Uniform light transmission	3.7.1	4.5.5.1	Visual
5. Percent light transmission	3.7.2	4.5.5.2	SME

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TABLE II - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 1.0%	Req't Par.	QA Par.	Test Method
6. Angular deviation	3.7.3	4.5.5.3	SME
7. Grade A, image quality	3.7.4.1	4.5.5.4.1	Device, Special Testing (DST)
8. Grade B, image quality	3.7.4.2	4.5.5.4.2	DST
9. Grade C, image quality	3.7.4.3	4.5.5.4.3	SME
MINOR: AQL 2.5%			
41. Identification marking	3.8	4.5.6	Visual

4.4 Inspection equipment.

4.4.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.4.2 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.5 Test methods and procedures.

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4.5.1 Storage temperature.- The filters shall be exposed to the temperatures indicated in 3.5.1 by means of a heating chamber and a cold test chamber capable of maintaining these temperatures until the filters under test are thermally stabilized. The filters shall then be allowed to return to standard ambient temperature until thermal stabilization is reached and then be tested to determine conformance to the requirements of 3.5.1.

4.5.2 Cleanliness.- Each filter of the applicable sample shall be visually examined for cleanliness in compliance with Specification MIL-O-13830.

4.5.3 Surface qualities.- Filters shall be examined for surface quality in compliance with Specification MIL-O-13830.

4.5.4 Cemented surfaces.- Filters to be inspected shall be viewed by a light source passing through the filter to the unaided eye. The maximum dimension of any discoloration, bubble or separation from the edge shall not exceed the tolerance specified in 3.6.

4.5.5 Optical characteristics.

4.5.5.1 Uniform light transmission.- The filter shall be examined visually with the unaided eye to the specified condition of 3.7.1 and shall meet the requirements specified therein.

4.5.5.2 Percent light transmission.- The filter shall be tested for percent light transmission by means of a spectrophotometer capable of obtaining transmission curves within a wavelength range from 400 to 700 millimicrons. The transmission curve obtained of the filter under test shall be compared with and must fall within the limits indicated on the applicable graph on the figure of those specified in 3.7.2.

4.5.5.3 Angular deviation.- The filter shall be tested for angular deviation by means of a testing device composed of a collimator and a viewing telescope provided with horizontal and vertical crosslines, and a medium for locating the filter under test. The collimator reticle shall be illuminated from the rear by an electric lamp. The testing device shall be provided with a scale radially arranged to read angular positions in minutes of arc. The collimator and telescope shall be adjusted until the crosslines coincide. The filter shall then be positioned between the two telescopes; any deviation shall be measured by reading the scale to determine conformance to the requirements specified in 3.7.3.

4.5.5.4 Image quality.

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4.5.5.4.1 Grade A.- Image quality of the filter shall be checked by means of an optical system as described in 3.7.4.1. The system is first checked for resolution and parallax without the filter and then with the filter in the system. Resolution and parallax shall not be impaired by the presence of the filter.

4.5.5.4.2 Grade B.- Image quality of the filter shall be checked with the aid of an eyepiece in an optical system as described in 3.7.4.2. The filter shall be tested by observing the image of a target without the filter and then with the filter in the system. The image quality shall not be impaired with the filter in the system.

4.5.5.4.3 Grade C.- Image quality of the filter is checked by means of a wall target and a three power (3X) dioptometer. The wall target can be constructed of four number 5 patterns of the National Bureau of Standards Test Charts arranged with the bars of the pattern thirty feet from the target and brought into sharp focus. The filter shall then be placed in front of the dioptometer, and any change in the sharpness of the focus shall not exceed the requirements of 3.7.4.3.

4.5.6 Identification marking.- Marking for identification shall be visually inspected for conformance in accordance with Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The filters shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619 and the applicable packaging data sheet. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The filter is intended to modify the spectral transmission of light by selective absorption.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class, and grade of filter required (see 1.2).
(c) Size of filter required if no drawing is provided (see 3.3).
(d) Provisions for production model including date of delivery (see 3.1).
(e) Selection of applicable levels of preservation, packaging and packing (see 5.1).
(f) Applicable stock number.
(g) Applicable packaging data sheet.

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6.3 Design information.- Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.4.2 Standard measuring equipment (SME).- Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.5 Cross reference to types, classes, and grades of Specification MIL-F-12716.

	MIL-F-12716A(Ord)	MIL-F-12716(Ord)
Types:	I II III	I III IV
Classes:	1 (Type III only) 2 (Type III only) 3 (Type III only)	D (Type IV only) E (Type IV only) F (Type IV only)
Grade:	A B C	B A C

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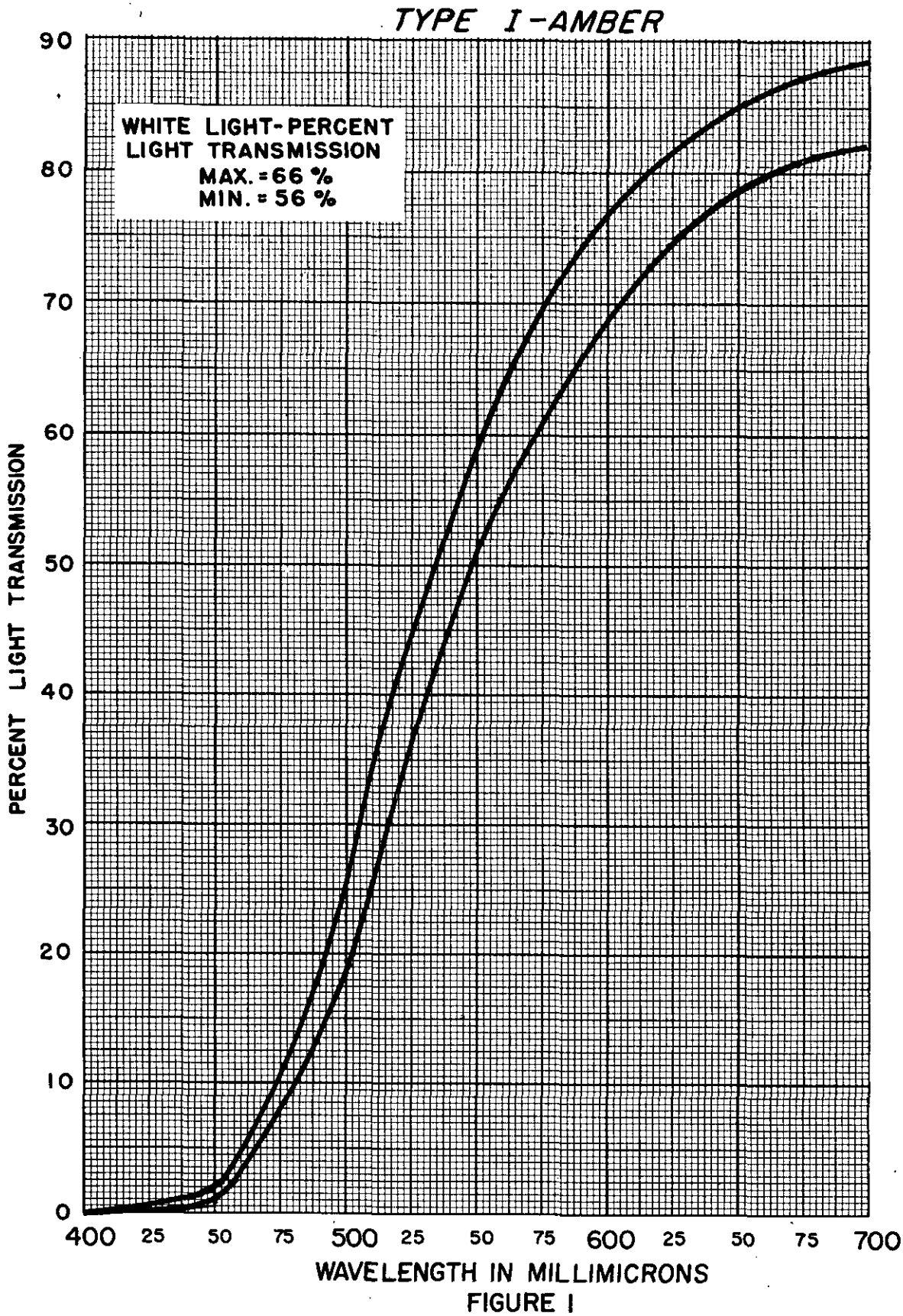


FIGURE I

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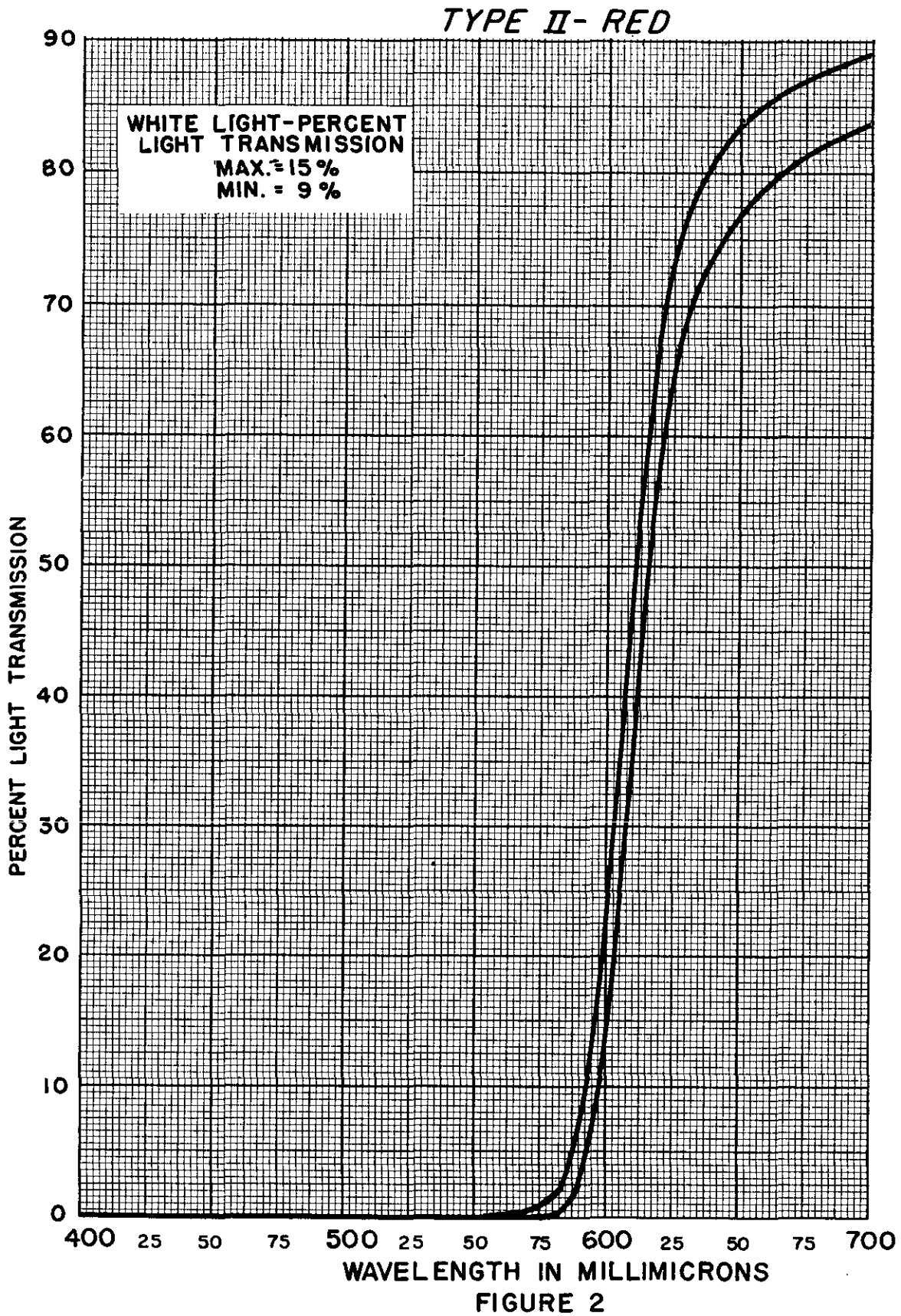


FIGURE 2

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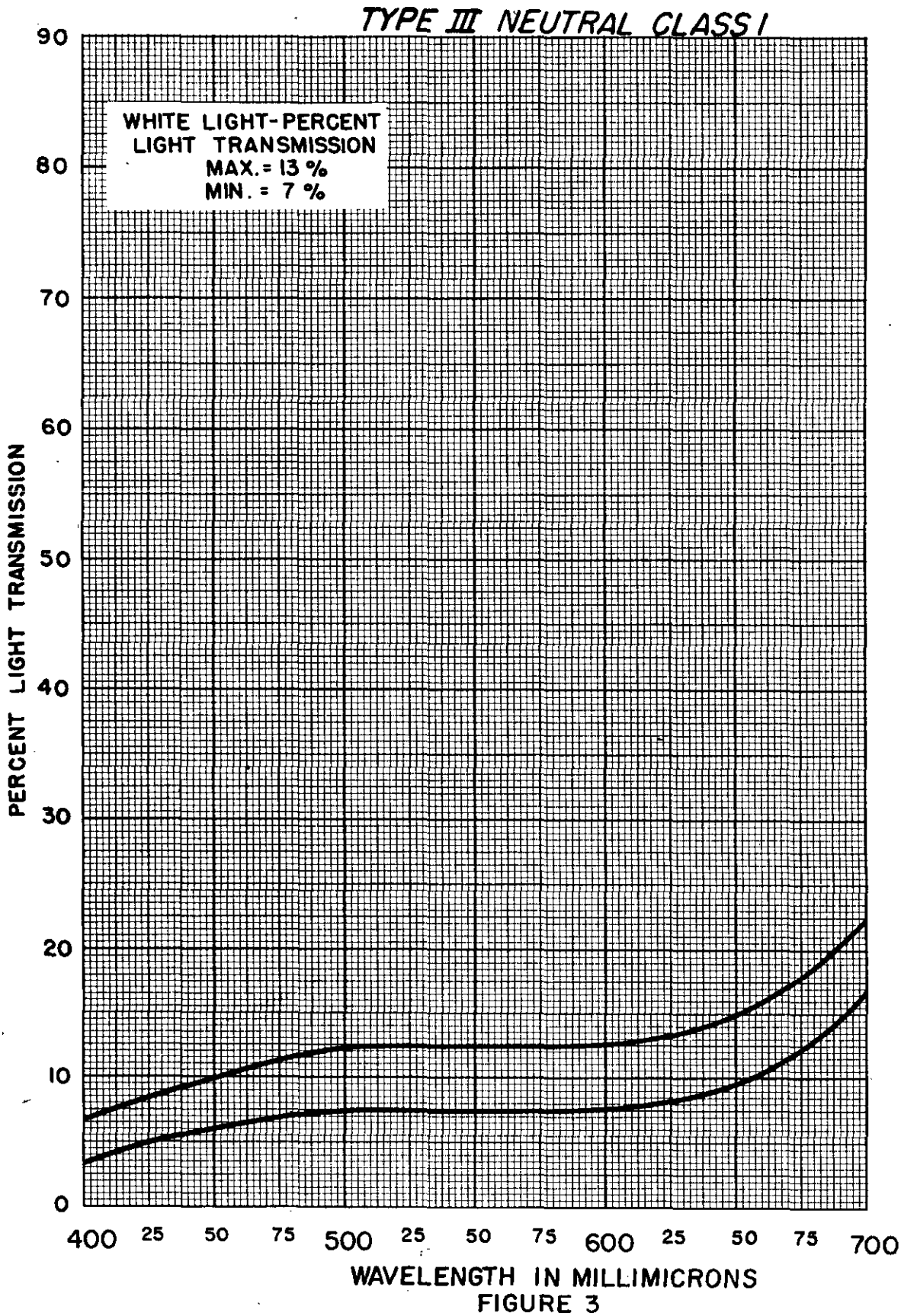


FIGURE 3

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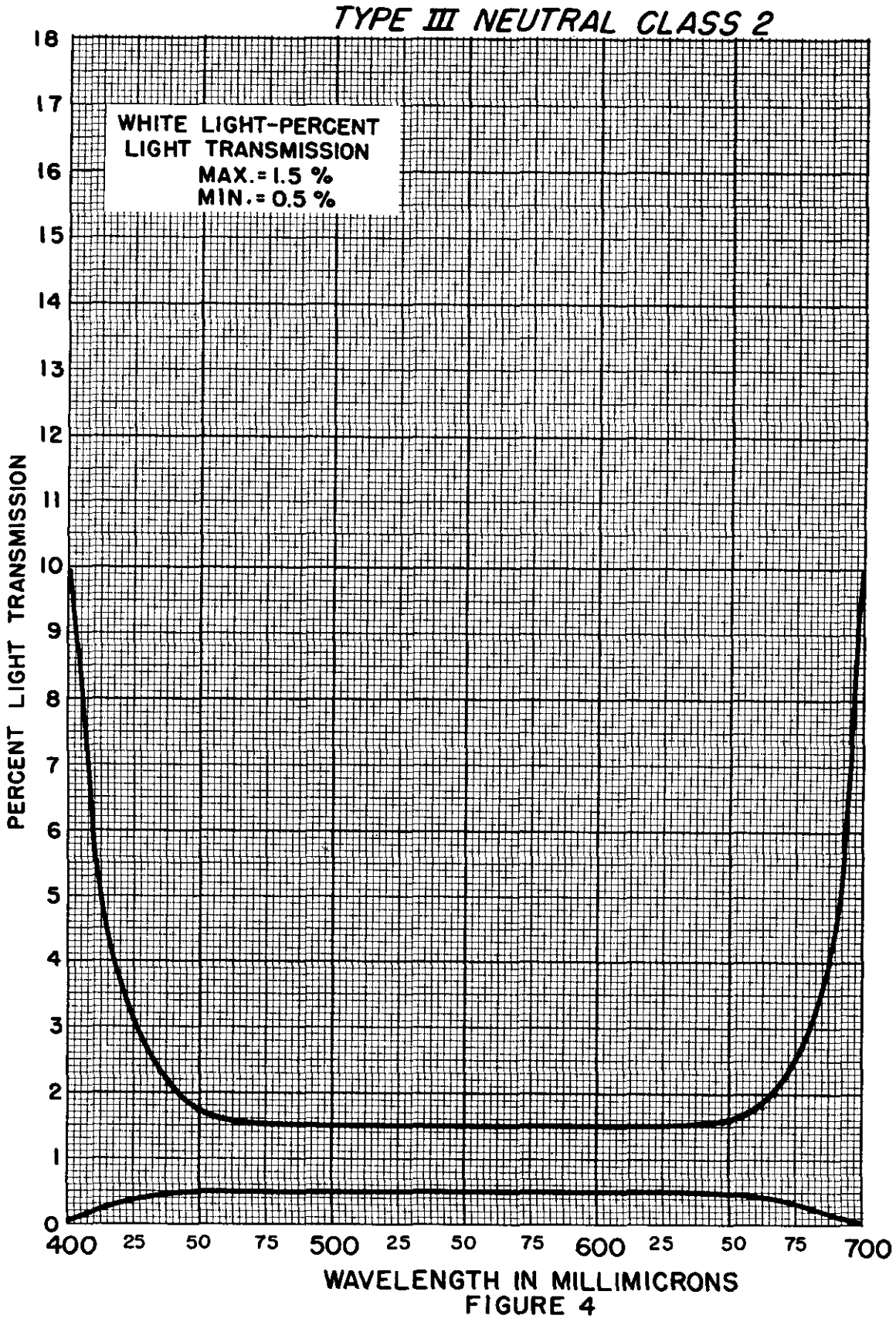


FIGURE 4

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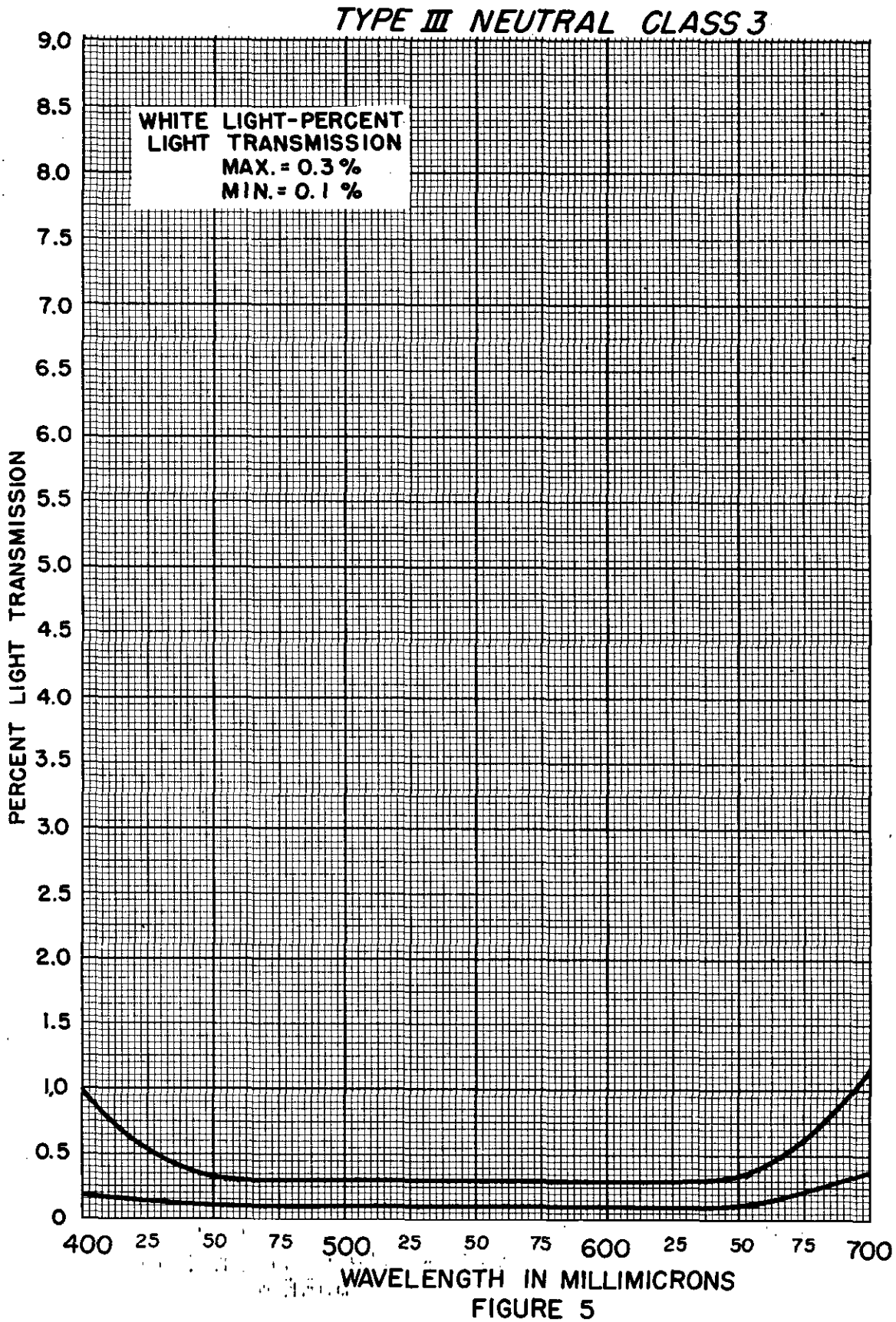


FIGURE 5

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